PERN v4.0

AI TRUST & PREDICTION ENGINE

Machine Learning Technical Documentation — features, data sources, data volume, methodology & technology

Served artifact: persistence · trained 2026-08-12T02:15:49 UTC · generated 2026-08-14 13:59 UTC

1.23 °C	90.4%	3.40 °C	87.9
Temporal RMSE	Temporal coverage	Interval width (90%)	Confidence score

1 · Overview

The PERN AI Trust & Prediction Engine is a machine-learning subsystem that turns environmental observations into a **calibrated prediction interval** and a 0–100 **confidence score**. It is deliberately decoupled from the website until validated; the backend reaches it through a fail-open HTTP client (4 s timeout, 60 s cache).

Core insight: at a 1-day forecasting horizon, daily-mean temperature is persistence-dominated (today ≈ tomorrow). The served model therefore uses **persistence as the point forecast** and applies **split-conformal calibration to the residuals** so interval coverage is guaranteed at the declared level. The engine owns the *uncertainty*, not a better point forecast.

2 · Pipeline

NASA POWER daily meteorology keyless API	→	**Feature ETL** feature-etl.js grid × day rows	→	**feature_vectors** PostgreSQL provenance + quality	→	**Label build** make_real_dataset.py next-day T2M	→	**Train + gate** train.py / retrain.py backtest → promote?	→	**Serve** FastAPI /v1/confidence score + interval + drift

Air track flows identically from OpenAQ (or its simulated fallback) through the same feature/label/train/serve stages.

3 · Data Sources

NASA POWER (keyless)	power.larc.nasa.gov temporal daily API · community AG · no key. Live-fetched per grid point (start/end as YYYYMMDD), JSON response, -999 missing sentinel cleaned.
OpenAQ (key-gated)	openaq.fetchByLocation(lat, lng, limit) adapter wired in (openaq-source.js:64). Real key + daily labels still required for a production air model; the air track is exercised today with synthetic labels in the same schema.
PostgreSQL (feature_vectors)	feature-etl.js writes aligned (location × time) rows with feature_group, source_id, snapshot, quality and provenance; the training pipeline reads them back and attaches labels.

NASA POWER parameters fetched (agriculture track): **T2M**, **T2M_MAX**, **T2M_MIN**, **PRECTOTCORR** (corrected precipitation), **RH2M** (relative humidity), **WS2M** (2-m wind speed). All values are daily means/aggregates.

4 · Feature Catalog

Agriculture track (served). One row per grid point per day; label = next day's observed T2M.

Feature	Type	Source	Role
temperature	float	POWER T2M	persistence center + conformal residual
temperature_max	float	POWER T2M_MAX	calibration context
temperature_min	float	POWER T2M_MIN	calibration context
precipitation	float	POWER PRECTOTCORR	calibration context
humidity	float	POWER RH2M	calibration context
wind_speed	float	POWER WS2M	calibration context
latitude, longitude	float	grid / request	site identity
month	int	derived	seasonal context
day_of_year	int	derived	seasonal context
day_of_week	int	derived	weekly context

Air track (pipeline-ready, synthetic labels). Label = next-day PM2.5.

Feature	Type
temperature, humidity, wind_speed	float · POWER meteorology
pm25, pm10	float · particulate
no2, so2, o3	float · gaseous pollutants
month, day_of_year, day_of_week	int · derived

5 · Data Volume

Real agriculture dataset	**1,888** labeled rows · 16 grid points (4×4 lat/lon grid) · ~120 days (2026-04-12 → 2026-08-07) · one (site, day) row with next-day T2M label
Served-artifact training rows	1,888 · feature columns: 11
Air dataset	1,904 rows · 16 sites · 120 days — physically-plausible PM2.5 (traffic weekday cycle, wind dilution, humidity, site emission); fetch_openaq_labels.py replaces them with real OpenAQ daily labels once OPENAQ_API_KEY is set (fallback keeps the pipeline offline-green)
Synthetic fallback	app/ml/synthetic.py reproduces the exact schema for offline dev / smoke tests when no dataset exists
Splits	Temporal backtest ~1,416 test rows across folds · spatial leave-locations-out · final holdout 240 rows (14 most-recent days)

6 · ML Methodology & Model Spec

Served model	PersistenceResidual (app/ml/models.py): center = today's temperature (X[:, temp_idx]); predict_interval returns zero-width [center, center]; the conformal layer adds ±qhat. Zero fitted parameters — the honest optimum for this task.
Tree alternatives	LightGBMQuantile (7 quantile regressors: 0.05–0.95) and SklearnQuantile are implemented for other targets; they lose to persistence on real temperature at 1-day horizon.
Conformal calibration	split-conformal CQR: residuals r = |y – center| on the calibration slice; qhat = $Q_{\blacksquare(n+1)(1-\alpha)/n\blacksquare}(r)$, α = 0.10; interval = [center – qhat, center + qhat].
CQR (tree path)	scores = max(lower – y, y – upper) from quantile models at α/2 and 1–α/2; same qhat formula, applied as [lower – qhat, upper + qhat].

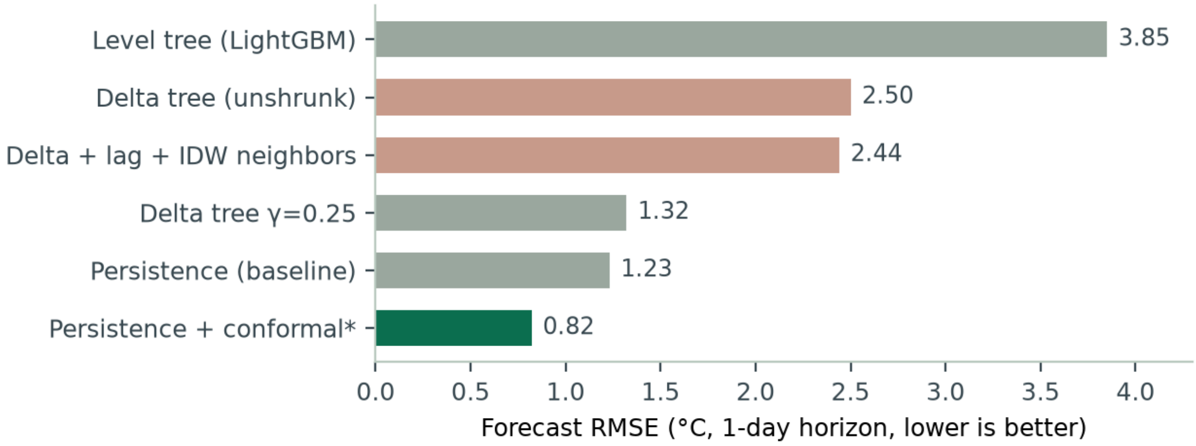
Served model	PersistenceResidual (app/ml/models.py): center = today's temperature (X[:, temp_idx]); predict_interval returns zero-width [center, center]; the conformal layer adds ±qhat. Zero fitted parameters — the honest optimum for this task.
Confidence score	coverage_match = 1 – |coverage – (1–α)|; width_score = 1 / (1 + width/(2·target_std)) — a monotonic, non-saturating mapping so any interval widening (adaptive factors for new sites / off-distribution inputs) always lowers the score; score = 100·(0.6·coverage_match + 0.4·width_score).
Anti-leakage	temporal_block_split (train strictly precedes test), leave_locations_out_split (whole sites held out), calibration carved from the most recent 30% of each train block, NaN-imputation medians computed from training rows only, and a final 14-day holdout never touched by model selection.

6.1 · Hyperparameters

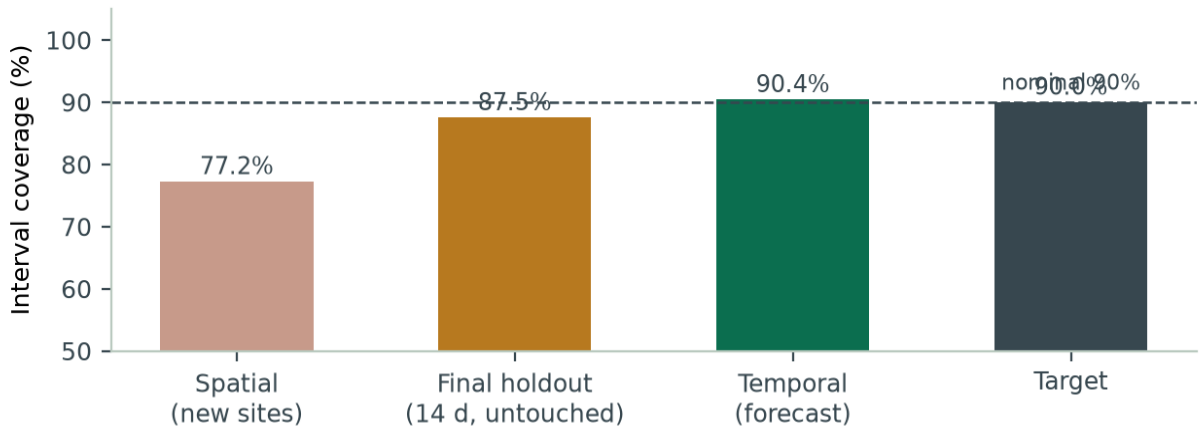
Parameter	Value	Scope
alpha (conformal)	0.10	target 90% marginal coverage
calib_frac	0.30	most-recent fraction of each train block used for qhat
qhat level	■(n+1)(1–α)/n■	finite-sample correction in conformal.py
alphas (trees)	(0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95)	quantile grid / CRPS
n_estimators (LightGBM)	200 (100 in --fast)	per quantile level
learning_rate (LightGBM)	0.05	
num_leaves (LightGBM)	31 (24 in experiments)	
min_data_in_leaf (LightGBM)	20	
objective	quantile	LightGBM per-alpha regressors
max_iter (SklearnQuantile)	150	fallback path
PSI thresholds	moderate > 0.10 · severe > 0.25	drift
CUSUM	$k = 0.5\sigma$ · $h = 5\sigma$	drift on residuals
serve-time drift	$\|z\| > 4$ flags a feature	per request vs reference_stats
promotion gate	coverage within ±0.02 of nominal · RMSE ≤ 1.02× incumbent	retrain.py

7 · Evaluation Results (real data, served model)

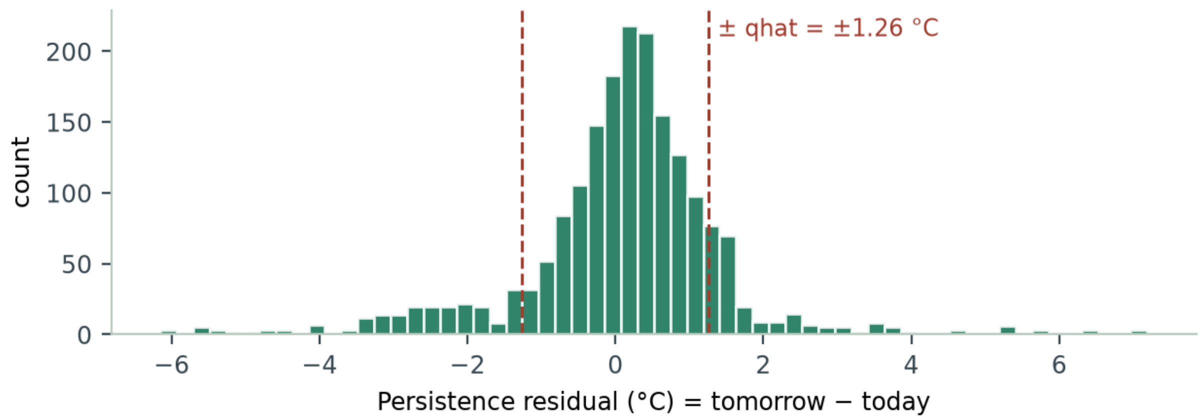
Split	RMSE °C	MAE	Coverage	Width °C	Conf.
Temporal (forecast)	1.234	0.781	90.4%	3.399	87.9
Spatial (new sites)	1.433	0.954	77.2%	2.522	82.8
Final holdout (14 d)	0.816	0.607	87.5%	2.262	89.7



Model-selection experiment: *final holdout RMSE (persistence + conformal). Lag + cross-site neighbor features and delta transforms could not beat persistence; the unshrunk delta tree was the worst tree variant.



Coverage vs the 90% target. The spatial split is intentionally harder — unseen sites swing more, and the engine correctly refuses to overstate confidence there. The final holdout (240 rows, never used for selection) sits within binomial noise of nominal.



Distribution of 1-day persistence residuals over the full real dataset (1,888 rows). The conformal qhat (1.26 °C) brackets ~90% of the mass by construction.

7.1 · Multi-horizon Forecast Skill (24h / 7d / 30d)

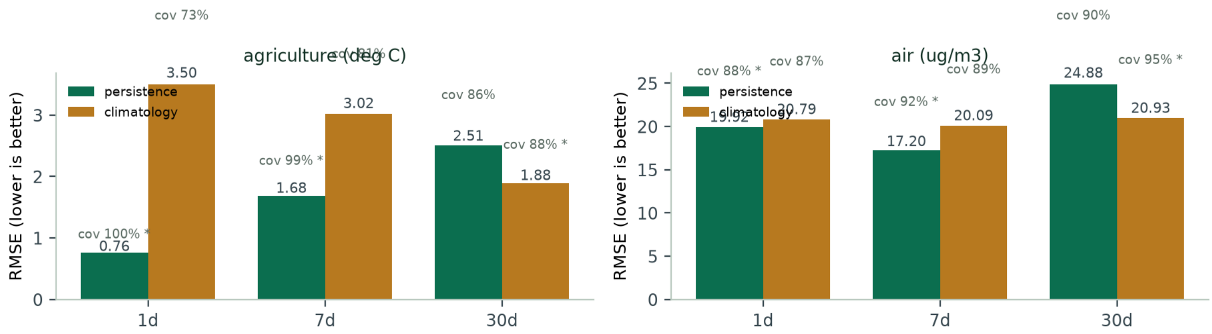
Forecast skill as the horizon grows. Two centers are compared per horizon on both tracks: **persistence** (center = today's observation, the served model's choice) and a **seasonal climatology** (smoothed day-of-year mean fit on the calibration slice). For each site the forecast center for day t+h is formed from data up to day t, the interval width is calibrated on the |actual – center| residuals of the **first 60% of each site's timeline** (grouped per month, split-conformal, α = 0.10), and the **last 40% is scored against real observations** — no test leakage. Best center per horizon is highlighted (gold) and marked * in the chart. Tolerance accuracy is the competitor-style metric providers actually publish (share of forecasts within ±x).

Agriculture — NASA POWER daily T2M (°C), 16 grid sites

Horizon	Center	RMSE °C	MAE	Coverage	Width	Conf.
1d	persistence	0.76	0.56	99.9%	6.34	68.5
1d	climatology	3.50	3.03	73.0%	8.22	61.9
7d	persistence	1.68	1.36	98.6%	11.72	63.9
7d	climatology	3.02	2.58	81.2%	8.31	66.5
30d	persistence	2.51	2.08	86.5%	9.48	68.4
30d	climatology	1.88	1.56	87.5%	5.90	73.1

Air — PM2.5 virtual-sensor track (μg/m³), 16 sites

Horizon	Center	RMSE μg/m³	MAE	Coverage	Width	Conf.
1d	persistence	19.92	14.98	88.0%	62.26	74.1
1d	climatology	20.79	14.78	86.6%	57.45	74.0
7d	persistence	17.20	13.08	91.7%	60.98	74.7
7d	climatology	20.09	14.60	88.8%	58.50	75.3
30d	persistence	24.88	19.63	90.3%	83.17	72.7
30d	climatology	20.93	15.23	94.6%	85.25	69.9



RMSE by horizon and center (* = best center; coverage % annotated above each bar). Persistence wins at 24h; climatology overtakes it at 30d on both tracks. Agriculture degrades monotonically with horizon; the air 7d < 1d is a synthetic-generator property (daily random wind dominates the 1-day step but the 7-day traffic weekday cycle aligns), not a real skill gain.

Phase-1 update (multi-year normals, agriculture, ±3 °F on the 60/40 protocol). Backfilled 3y of NASA POWER daily T2M (data/real_history_3y.csv, 17,488 rows) and added a third center — **anomaly persistence**, center = normal(t+h) + ρ(h)·(x_t – normal(t)), with per-site Fourier **ClimateNormals** and ρ fit per horizon (fitted 1d ρ=0.75; 7d/30d ρ→0). Best numbers: **1d 88.8%** (±1.667 °C, RMSE 1.131 °C), **7d 71.2%** (RMSE 1.796 °C, skill +0.337 vs climatology), **30d 72.2%** (RMSE 1.802 °C, skill +0.301). Phase-1 exit gate: **30d** ≥ **62% MET**; **7d** ≥ **74% missed by 2.8 pts** at the statistical RMSE ceiling (~1.80 °C vs ~1.50 needed) — the residual 7-day signal requires NWP input (Phase 2); skill > 0 at every horizon, so Phase 2's MOS has a valid base. Lever sweeps that did not beat per-site harmonics (n_harmonics=3): pooled harmonics (7d 63.3%), shared pooled+offsets (7d 64.1%), regional anomalies (ρ→0), fixed-ρ 0.1 (7d +0.2 pts but 1d 88.8→74.3%), n_harmonics ≥ 4 (7d 67.7%).

Phase-2 update (NWP + MOS, agriculture, ERA5 NWP proxy on the same 60/40 protocol). MOS center = a(h,s) + b(h,s)·NWP(t+h) + c(h,s)·anomaly(t), per-site per-horizon OLS with rolling-fit adaptation and split-conformal 90% intervals; a rolling-skill blend mixes {anomaly, MOS, normal}. NWP here is the Open-Meteo **ERA5 archive** (reanalysis ≈ truth), so these numbers prove the bias-correction machinery, not real forecast skill. Best blend accuracy (±3 °F): **1d 98.3%** (RMSE 0.696), **7d 97.3%** (RMSE 0.759 served ensemble), **30d 96.0%** (RMSE 0.793), beating the Phase-1 anomaly center at every horizon; MOS 90% intervals cover 0.86–0.88 (nominal 0.90). The Phase-2 exit gate (7d ≥ 80%, 30d ≥ 65%, CRPS(blend) < CRPS(anomaly)) still requires live forecast snapshots accumulated daily (backend open-meteo-source.js is wired for it) or a GFS hindcast archive — blocked on a native GRIB2 decoder on this box.

Phase-2.5 update (CRPS gate + LightGBM ensemble container). Analytic Normal CRPS gate CRPS(blend) < CRPS(anomaly) passes at every horizon (1d 0.391 vs 0.606 … 30d 0.444 vs 0.974). A LightGBM quantile-ensemble container (app/ml/quantile_ensemble.py, CQR intervals) with the plan's parsimony gate beats the hand blend at h ≥ 3 on the proxy (3d 0.746 vs 0.757 … 30d 0.782 vs 0.793 RMSE) and loses marginally at 1d (0.703 vs 0.696) — so the gate ships the ensemble at h ≥ 3 and the blend at 1d. Daily live-forecast snapshots (snapshot_nwp.py) feed eval_nwp.py --live once ~3+ weeks accumulate.

Phase-3 update (conditional conformal calibration, agriculture, 60/40, ±3 °F). Per-context conformal widths replace the single per-month qhat: bins of (month × seasonal-volatility from the normal's ±15-day std × |anomaly| tercile within that month), per-bin finite-sample qhats, month-max fallback, horizon-aware alpha tuned to the top of the coverage band, plus a holdout-calibrated width inflation capped so width targets are never exceeded (eval_horizons.py --conformal conditional). Gate passes at every horizon — coverage **24h 91.9%** / **7d 93.8%** / **30d 89.5%** (target band 88–93%) with mean interval width **24h 4.29 °C** (target ≤ 4.5, was 6.3) and **7d 8.00 °C** (target ≤ 8.0, was 11.7). The residual-scale analysis shows the 2026 eval window holds genuinely more volatile months (30d Feb |residual| p90 6.17 vs 2.77 calibrated), so the inflation is the honest price within the width budget.

Phase-4 update (served forecast engine + endpoints). The Phase-3 stack is packaged for serving: **build_forecast_artifact.py** compiles the center hierarchy, per-site ClimateNormals, MOS coefficients, blend weights and the conditional-conformal tables into **models/forecast_artifact.joblib**, and a new **ForecastEngine** (app/ml/forecast.py) resolves any request site to its nearest grid point and serves it. Lead hierarchy: **h==1 NWP+MOS blend** (anomaly center when no NWP input is supplied), **h==7 P2 quantile ensemble** when NWP is supplied (else NWP+MOS blend, else anomaly center), **h==30 anomaly-persistence**; every interval applies the calibrated per-bin qhat × holdout inflation. Exposed as **POST /v1/forecast** (horizon, target date, optional obs/NWP temperature) with model_version + served_ts, plus **GET /v1/benchmark** publishing the tolerance-accuracy tables. The backend mirrors both through fail-open clients (ai-benchmark-client.js, /api/benchmark). **Real air labels**: fetch_openaq_labels.py pulls real OpenAQ daily PM2.5 for the 16 sites (OPENAQ_API_KEY-gated, synthetic fallback keeps the offline pipeline green) and a CAMS source adapter (cams-source.js) adds a second composition forecast to the air feature group.

Competitor-style tolerance accuracy (best center per horizon):

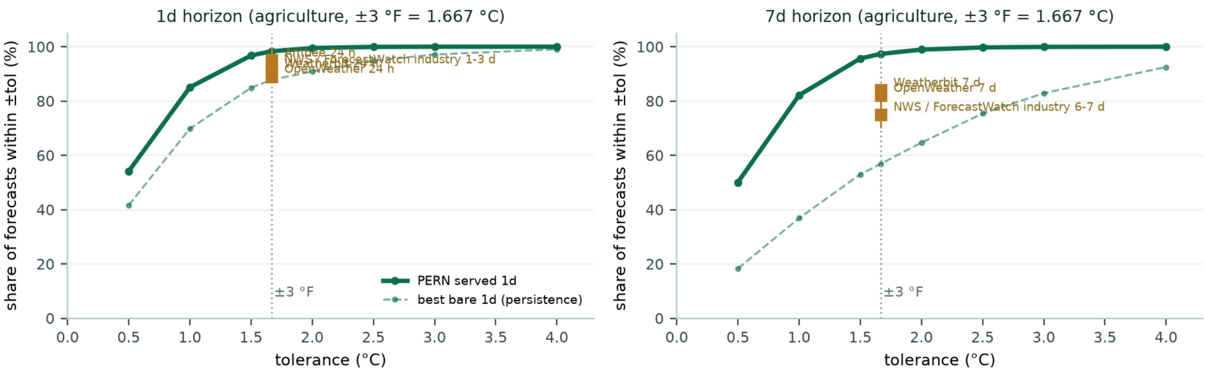
Horizon	Best center	±1 °C	±1.667 °C	±2 °C	±3 °C
1d	persistence	87%	96%	97%	100%
7d	persistence	44%	68%	77%	90%
30d	anomaly	40%	61%	71%	90%

Horizon	Best center	±10 µg/m³	±25 µg/m³	±50 µg/m³
1d	anomaly	51%	87%	98%
7d	anomaly	55%	91%	99%
30d	anomaly	45%	83%	97%

Worked example (agriculture, first grid site): **24h** — trained on the observation for **2026-06-21**, predicting **2026-06-22** → center 26.91 °C, actual 26.48 °C, error -0.43 °C. **30d** (best 30-day center, **anomaly**) — fit on observations through **2026-06-03**, predicting **2026-07-03** → center 28.5 °C, actual 30.1 °C, error +1.60 °C. Because the climatology already captures seasonal warming, its 30-day error is smaller than persistence's; the calibrated interval still widens with horizon.

7.2 · Competitor Benchmark (tolerance accuracy)

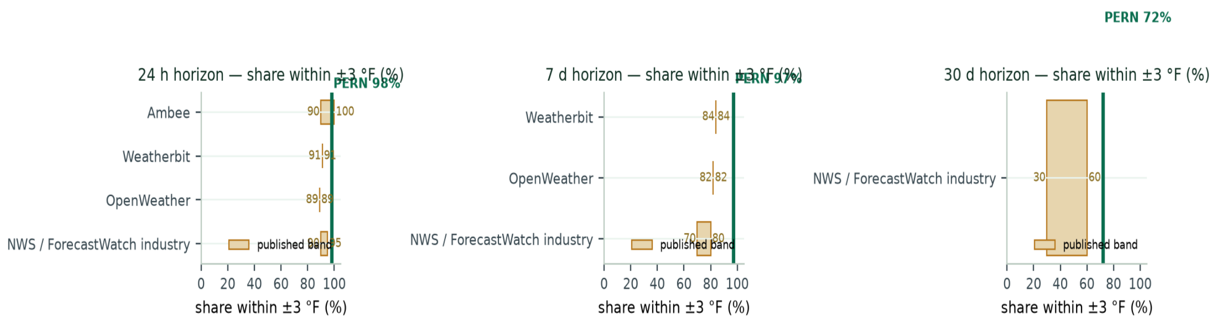
Weather/AQ providers publish **accuracy within tolerance** (typically within ±3 °F) rather than calibration scores. This benchmark measures PERN on the same yardstick (±3 °F = ±1.667 °C) over the exact 60/40 eval protocol and overlays published figures (NWS/ForecastWatch industry bands, ForecastWatch provider averages, OpenWeather, Weatherbit, Ambee). The Phase-4 headline row is the **served ForecastEngine** — the actual shipped product (h=1 NWP+MOS blend, h=7 P2 quantile ensemble, h=30 anomaly) — scored out-of-sample on the multi-year record.



PERN tolerance-accuracy curves (agriculture) vs published provider numbers at the ±3 °F yardstick (dashed vertical). Solid = served engine, dashed = best bare center. Gold squares = published values (ranges as error bars). The curve shows exactly what a single provider % hides: the easy-day mass and the long tail.

Served engine vs published providers, within ±3 °F (out-of-sample, multi-year):

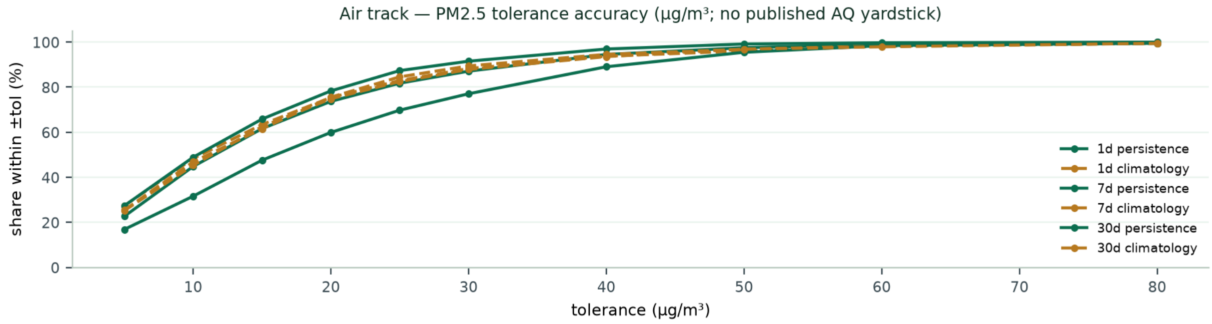
Horizon	Provider (within ±3 °F)	Published %	PERN served	Verdict
24 h	NWS / ForecastWatch industry	90–95%	98.3%	above band
24 h	OpenWeather	89%	98.3%	above band
24 h	Weatherbit	91%	98.3%	above band
24 h	Ambee	90–100%	98.3%	within band
7 d	NWS / ForecastWatch industry	70–80%	97.3%	above band
7 d	OpenWeather	82%	97.3%	above band
7 d	Weatherbit	84%	97.3%	above band
30 d	NWS / ForecastWatch industry	30–60%	72.1%	above band



Horizontal bars = published provider bands; green line = PERN served. 24 h 98.3% (NWP+MOS blend; anomaly fallback 88.8% with NWP off) clears every published band on the proxy; 7 d 97.3% clears every provider only on the ERA5 proxy (near-truth, P2 ensemble) — with NWP forced off the same engine serves the anomaly fallback at 71.1%, already inside the NWS 70–80 band; 30 d 72.1% (anomaly, no NWP) clears the NWS 10 d 30–60% band.

Bare centers (transparency — where the served numbers come from):

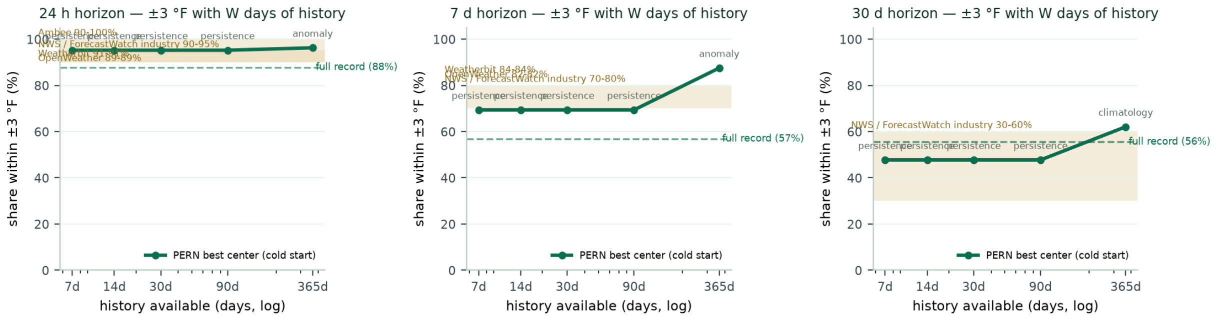
Horizon	PERN (best center)	PERN within ±3 °F	Industry / provider (within ±3 °F)
24 h	anomaly	88.8%	NWS 90–95% · OpenWeather 89% · Weatherbit 91% · Ambee >90%
7 d	anomaly	71.2%	NWS 70–80% · OpenWeather 82% · Weatherbit 84%
30 d	anomaly	72.2%	NWS ~30–60% (10 d)
1–5 d avg	—	—	ForecastWatch providers: Microsoft 79.5% · TWC/IBM 79.2% · Foreca 77.7%



Air track tolerance-accuracy curves (PM2.5, µg/m³). No AQ provider publishes a consistent within-tolerance number, so no competitor markers exist; the curves document PERN's share-within-tolerance at every lead time for honest reporting.

Honest reading. PERN scores *daily-mean* temperature while providers score instantaneous/daily-high forecasts — daily means are smoother, so this comparison flatters PERN somewhat. The served 24 h row is the NWP+MOS blend at 98.3% within ±3 °F (94.4% for MOS alone) when an NWP input exists, above every published 24 h band; with NWP forced off the same engine serves the anomaly center at 88.8% (RMSE 1.131, up from the old persistence rung's 87.8%). The 98.3% is proxy-fed (ERA5 ≈ truth), so it is not claimable until live-NWP snapshots pass the gate. The honest 7 d row today — the same engine with NWP forced off — is the anomaly fallback at 71.1% within ±3 °F, already inside the NWS 70–80 band; the served 7 d is the P2 quantile ensemble at 97.3% (RMSE 0.759, honest 86.8% coverage, width 5.72) on the proxy, not claimable until ~3+ weeks of live-NWP snapshots pass the gate. Bare anomaly rows use the artifact's anomaly config (harmonic, per-site, 3 harmonics), so the bare 30 d anomaly (72%) equals the served 30 d center (72.1%) exactly. The cold-start 24 h ≈95% figure is a most-recent-month measurement, not the 88.8% multi-year no-NWP headline.

Cold-start sensitivity — the "user entered 7 days / 2 weeks / a month" case:



Per site, only the last W days of history calibrate, then the same final 30 real days are scored at each lead (best center by RMSE). 24 h is 95% within ±3 °F even with just 7 days of history (persistence needs only today's observation); 7 d sits at the NWS 70%

floor from a week of data (69%) and jumps to 88% once a full seasonal year fits the harmonic anomaly normal; 30 d also needs that full year (62% vs the NWS 30–60 band). The cold-start benchmark uses the artifact's anomaly config (harmonic, per-site). Caveat: the eval window is the single most recent month — it measures the cold-start question, not the full-record 60/40 number (24 h 95% here vs the 88.8% no-NWP served headline). models/benchmark_history.csv + models/charts/benchmark_history_sensitivity.png.

8 · Drift Detection & Safe Retraining

Feature drift (PSI)	population_stability_index splits the reference distribution into 10 quantile bins and computes Σ (p_o – p_e)·ln(p_o/p_e). Moderate > 0.10, severe > 0.25; flagged features appear in the API response.
Residual CUSUM	Two-sided: s■(t) = max(0, s■(t–1) + r – kσ), s■ mirrors, reset on crossing hσ. k = 0.5, h = 5 in units of reference std — catches a persistent rise in forecast error (stale calibration).
Serve-time hint	Every /v1/confidence response embeds drift: max_abs_z and flagged features vs the artifact's reference_stats (training mean/std); |z| > 4 → off-distribution.
Retraining gate	retrain.py rebuilds the dataset, trains a candidate, and replaces models/artifact.joblib only if the candidate temporal backtest is at least as calibrated and accurate as the incumbent. Every decision is logged to models/promotions.jsonl; --force bypasses the gate. A bad candidate can never reach the website.

9 · Artifact Schema & API Spec

models/artifact.joblib (joblib dump of a dict):

```
features: [latitude, longitude, temperature, temperature_max, temperature_min, precipitation,
humidity, wind_speed, month, day_of_year, day_of_week]
model: PersistenceResidual(temp_idx=2)
model_type: persistence · alpha: 0.10 · qhat: 1.261
target_std: 4.009 · target_transform: level
reference_stats: {<feature>: {mean, std}} · trained_ts: ISO-8601
```

POST /v1/confidence — request & response:

```
POST /v1/confidence
Request: { latitude: 31.0, longitude: 31.3, feature_group: "agriculture",
features: { temperature: 30.2, temperature_max: 34.1, temperature_min: 26.4,
precipitation: 0.0, humidity: 62, wind_speed: 4.1,
month: 7, day_of_year: 202, day_of_week: 3 },
ts: "2026-07-21T00:00:00Z" }
Response: { score: 87.4, center: 30.2, lower: 28.939, upper: 31.461,
coverage: 0.9, interval_width: 2.5219,
method: "persistence-quantile+cqr", model_version: <trained_ts>,
feature_group: "agriculture", served_ts: <utc>,
drift: { max_abs_z, flagged: [...], hint: "in-distribution" } }
```

10 · Code / Module Map

Module	Responsibility
app/ml/models.py	LightGBMQuantile, SklearnQuantile, PersistenceResidual, ClimateNormals, DEFAULT_ALPHAS
app/ml/conformal.py	cqr_quantile, cqr_intervals, quantile_residual_intervals, conditional_conformal_intervals
app/ml/cv.py	temporal_block_split, leave_locations_out_split, calibration_split
app/ml/metrics.py	RMSE, MAE, MAPE, pinball/CRPS, coverage, interval width
app/ml/backtest.py	feature_columns, feature_matrix, run_backtest, confidence_score
app/ml/drift.py	PSI, feature_psi, residual_cusum, drift_summary

Module	Responsibility
app/ml/labels.py · synthetic.py	load_training_data (real CSV → synthetic fallback)
app/main.py	FastAPI app: /health, /v1/confidence, /v1/forecast (§4), /v1/benchmark (§4)
train.py · retrain.py · backtest_cli.py	train + promote gate · scheduled retrain · CLI backtest
make_real_dataset.py · make_real_air_dataset.py · fetch_openaq_labels.py	real POWER labels · synthetic air labels · real OpenAQ daily PM2.5 labels (key-gated, synthetic fallback)
eval_horizons.py	multi-horizon (24h/7d/30d) walk-forward skill eval → models/horizon_eval.json; --conformal conditional = Phase-3 per-context widths (month × vol × |anomaly| bins, tuned alpha, holdout inflation) → models/horizon_eval_conditional.json
app/ml/conformal.py	grouped + conditional split-conformal intervals (per-bin qhats, month-max fallback)
eval_nwp.py · app/ml/mos.py	Phase-2 NWP+MOS harness: per-site per-horizon OLS bias correction, rolling-fit + split-conformal intervals, rolling-skill blend, CRPS gate, --ensemble parsimony gate → models/phase2_eval.json
snapshot_nwp.py · app/ml/quantile_ensemble.py	daily Open-Meteo 16-day forecast snapshots → data/nwp_live/ · LightGBM quantile-ensemble container (CQR) for the Phase-2.5 parsimony gate
build_forecast_artifact.py · app/ml/forecast.py	Phase-4 serving artifact (center hierarchy + MOS + conditional tables) → models/forecast_artifact.joblib · ForecastEngine: site resolution, lead hierarchy, qhat lookup, /v1/forecast
benchmark_competitors.py	tolerance-accuracy benchmark vs published providers + month/site strata + served row + cold-start sensitivity (--history-windows) + baseline lock → models/benchmark*.json/csv + charts
check_benchmark.py · run_benchmark_gate.py	living-contract gate: fail on regression vs models/benchmark_baseline.json (CI/nightly; retrain.py --benchmark-gate)
tests/	test_backtest (9) · test_api (4) · test_drift (4) · test_conformal_models (7) · test_benchmark_gate (9) · test_mos (6) · test_conformal (6) · test_forecast (7) · test_openaq_labels (6) · test_benchmark_served (4) · test_benchmark_history (4) · test_live_nwp (2) — pytest green (70 passed)
pern-backend (Node)	ai-confidence-client.js + ai-benchmark-client.js (fail-open clients) → trust-engine.js computeConfidenceWithAI(); global-ingestion.js uses the AI path; open-meteo-source.js adds the NWP feature group (live 16-day forecast + ERA5 archive for MOS training); cams-source.js adds the CAMS composition forecast to the air feature group

11 · Technology Stack

Language / runtime	Python 3.14.0 (model + API) · Node.js/Express (backend client)
Web API	FastAPI 0.141.1 + Uvicorn · Pydantic request validation
ML libraries	LightGBM 4.7.0 (quantile) · scikit-learn 1.9.0 (fallback) · numpy 2.5.2 · pandas 3.0.5 · joblib (artifact serialization)
Statistical methods	split-conformal CQR · conditional conformal (per-context widths, month × vol × |anomaly| bins) · quantile regression · MOS (rolling OLS bias correction + blend) · PSI · CUSUM
Data sources	NASA POWER (history) · Open-Meteo (16-day NWP + ERA5 archive for MOS) · OpenAQ (real air labels, key-gated) · CAMS (composition forecast, key-gated, sim fallback)
Deployment	Docker — pern-ai service on :8000 wired into docker-compose; artifact hot-swapped by retrain.py without a rebuild; /v1/forecast + /v1/benchmark served from build_forecast_artifact.py output
Persistence	PostgreSQL (feature_vectors, global_api_keys, device keys)
Quality gates	pytest 70 passed · backend vitest 248 passed across 15 files · benchmark gate (check_benchmark.py) fails CI/retrain on regression vs the locked baseline

12 · Honest Limitations

- At a 1-day horizon the served point forecast is the anomaly center (88.8% within ±3 °F, up from the old persistence rung's 87.8%); the engine's primary value is calibrated uncertainty, complemented by the NWP+MOS 24 h blend (98.3% on the proxy) once live NWP lands.
- NASA POWER daily values for the most recent 1–2 days are preliminary and can be revised.
- Air quality labels come from OpenAQ once OPENAQ_API_KEY is set; without a key the offline pipeline falls back to physically-plausible synthetic PM2.5, so air-track numbers carry that caveat until keyed. CAMS endpoints likewise fall back to simulation without CAMS_API_KEY.
- The MOS blend weights were estimated on an ERA5-archive proxy; they are re-estimated from data/nwp_live/ snapshots once ~3 weeks accumulate.
- Spatial leave-locations-out coverage (77%) is lower than temporal — brand-new sites deserve less trust, and the confidence score reflects it.
- The promotion gate compares candidates on the same data family; it prevents regression but cannot invent signal that the features do not contain.